

Bellabeat User Activity & Health Insights

DATA ANALYSIS REPORT

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1.0 Business Task

Title: Smart Device Usage Analysis for Bellabeat

Business Context:

Bellabeat, a high-tech manufacturer of health-focused smart products for women, aims to expand their presence within the global smart device market. To achieve this goal, Bellabeat seeks to understand how consumers currently use smart devices to monitor and improve their health. These insights will guide marketing strategies for Bellabeat's flagship product, *Leaf*.

Objective:

The project aims to identify which key health metrics – such as sleep, activity levels or calorie expenditure – most strongly correlating with improved overall health.

Key Questions:

- What are some trends in smart devices usage?
- How could these trends apply to Bellabeat customers?
- How could these trends help influence Bellabeat marketing strategy?

Data Scope:

The analysis uses *Fitbit Fitness Tracker Data* by Möbius, covering sleep, activity and weight metrics collected between **March 12, 2016**, and **May 12, 2016**.

Deliverables:

1. Detailed executive report summarising findings, insights, and actionable recommendations for Bellabeat's marketing team.

2.0 Data Sources

This project uses the publicly available **FitBit Fitness Tracker Dataset** by Mobius to perform its analysis and construct applicable recommendations for Bellabeat. The dataset contains multiple data files pertaining to various user health metrics collected over the 90-day period. Within the scope of this project, only a selected subset of files is relevant to the analysis:

1. **daily_activity.csv**

- Contains each participant's daily activity data, including total steps, total distance, active minutes etc.
- Used to analyse general activity levels and its relationship to weight and calorie expenditure.

2. **heartrate_seconds.csv**

- Contains heartrate data of each participant, measured every second.
- Used to analyse heartrate patterns and its relationship to weight.

3. **hourly_calories.csv**

- Contains calorie expenditure data for each participant, measured every hour.
- Used to analyse calorie expenditure and its relationship to weight.

4. **hourly_intensities.csv**

- Contains activity intensities data for each participant, measured every hour.
- Used to analyse intensities patterns and its relationship to weight.

5. **minute_mets.csv**

- Contains MET (Metabolic Equivalent) data for each participant, measured every hour.
- Used to analyse MET's patterns and its relationship to weight.

6. **sleep_day.csv**

- Contains each participant's daily sleep duration, including minutes asleep and minutes in bed, measured daily.
- Used to analyse general sleep patterns and its relationship to weight.

7. **weight_log_info.csv**

- Contains each participant's weight data, including BMI and fat percentages, measured daily.
- Used to analyse changes in weight and as the dependent variable in weight - correlation analysis.

Each data file was imported and cleaned using Excel or MySQL to prepare for analysis. Final exploratory, statistical, and visual analyses were performed using Tableau.

3.0 Data Cleaning & Preparation

3.1 Data Consolidation

The initial step in cleaning the **FitBit Fitness Tracker Dataset** by Mobius for analysis involved addressing the organisation of the raw data files. Upon downloading, the zip file was found to contain two separate directories: *Fitabase Data 3.12.16-4.11.16* and *Fitabase Data 4.12.16-5.12.16*. Each folder stored datasets that corresponded to specific date ranges, resulting in the relevant data being split between the two directories. To ensure a seamless cleaning process, it was necessary to first merge each related files into a single file.

This was accomplished by developing a Python script, *merge_files.py*. By consolidating the datasets, subsequent data cleaning and analysis could be performed more efficiently and accurately.

3.2 Data Cleaning

Prior to analysis, each datafile underwent a comprehensive cleaning and transformative procedure using both Excel and MySQL. MySQL was utilised for handling larger files that surpassed Excel's capacity. The process involved several key steps to ensure integrity and accuracy of the data.

1. **Removal of Duplicates:** Duplicate records were identified and deleted to maintain the accuracy of results and subsequently prevent bias analysis.
2. **Time and Date Formatting Corrections:** Date formats were standardised to the conventional Australian format.
3. **Column Splitting:** Fields such as 'activityMinute' originally consisting of both date and time information, were divided into distinct time and date columns to enable more precise time-series analysis.
4. **Standardisation of Units and Formats:** Units of measurement, such as distance and weight were standardised and significant figures were aligned.
5. **Handling Missing or Null Values:** Instances of missing data were carefully identified and managed.
6. **Data Type Conversions:** Text fields were converted to numeric or datetime types as necessary, enabling calculations and aggregation to be performed during analysis.
7. **Standardisation of Column Names:** Updating ambiguous or inconsistent field names to ensure reliable grouping and analysis.

**Complete cleaning log can be found in 'cleaning_log.xlsx'*

3.3 Data Limitations

A key limitation of this project is the substantial volume of missing data across all data files. This likely affected the capacity to perform certain analyses and thus may have influenced the representativeness of the findings. As a result, the conclusions should be interpreted with caution.

4.0 Analysis Summary

4.1 Trends in Smart Device Usage

The 90-day analysis indicated no significant improvements in overall health among users, as measured by health metrics such as weight loss, BMI, and resting heart rate. However, a comparison between healthier and less healthy users revealed several notable trends:

- Users with a lower body weight take more steps and were generally more active than those with a higher body weight. [add stat here]
- Users with a lower body weight slept more minutes on average than those with a higher body weight. [add stat here]
- Users with a higher body weight had a greater daily calorie intake than those with a lesser body weight. [add stat here]
- Users with a higher body weight had fewer active days than those with a lesser body weight. [add stat here]

4.2 Application of Trends to Bellabeat Customers

These observed trends can help Bellabeat better anticipate how different customer demographics use its products. Understanding differences among user segments can inform the development of tailored features designed to increase engagement across the user base. For example, segment-specific functionalities could motivate less active users or reward healthier behaviours, thereby addressing the varying needs and habits of Bellabeat's diverse customers. Collectively, these improvements may increase user satisfaction and support growth in overall user adoption.

5.0 Visualisation & Key Insights

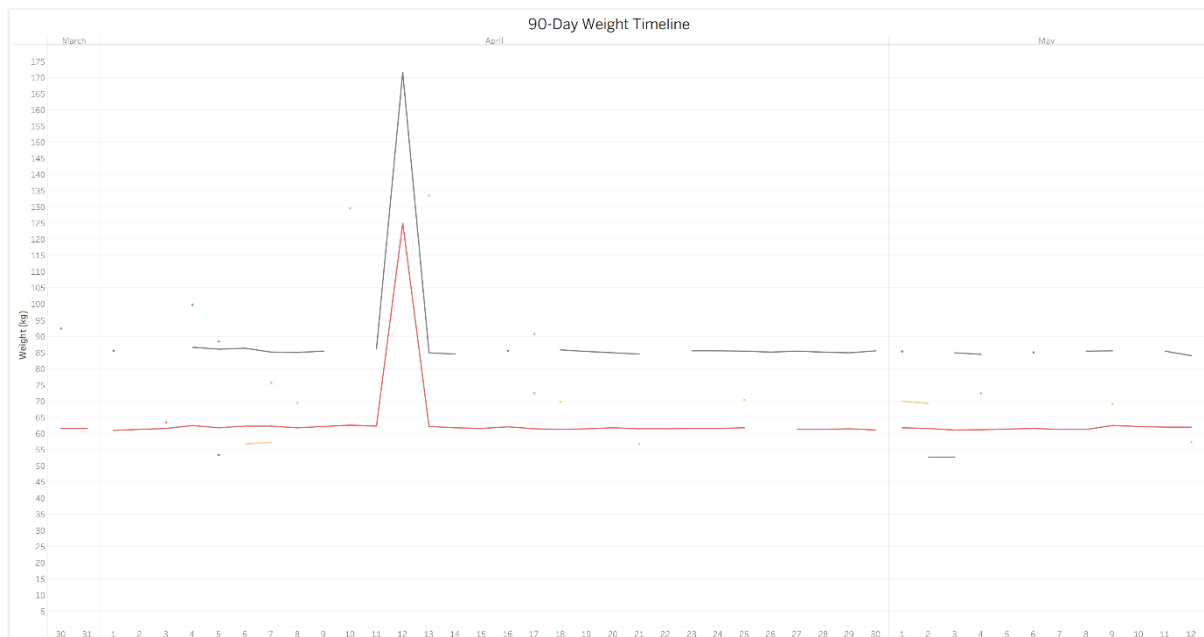


Figure 1: Weight Timeline

“90-day weight timeline” illustrates the variations in weight over the 90-day period for all users, revealing a steady flat linear trend. Indicating, apart from the anomalies, no significant fluctuation in body weight during the analysis timeframe.

Implication

The observed stability in user weight over the 90-day period is unexpected given that these individuals are users of a fitness product. Depending on individual fitness goals, a gradual increase or decrease in body weight would typically be anticipated over a period of this length; however, no such trend is evident.

Further investigation is therefore warranted to better understand user motivations for purchasing fitness products in the absence of measurable health improvements. One possible explanation is that some consumers use these products for aesthetic reasons or to reinforce a fitness-oriented self-identity rather than to achieve specific health outcomes.

Anomalies

Notably, two users (IDs 6922181067 and 8877689391) exhibited an anomalous spike in recorded weight on 12 April, with values approximately doubling before returning to baseline within a single day. Such rapid and extreme fluctuations are not physiologically plausible and are indicative of a data entry or measurement error. Accordingly, these observations should be excluded from the overall analysis.

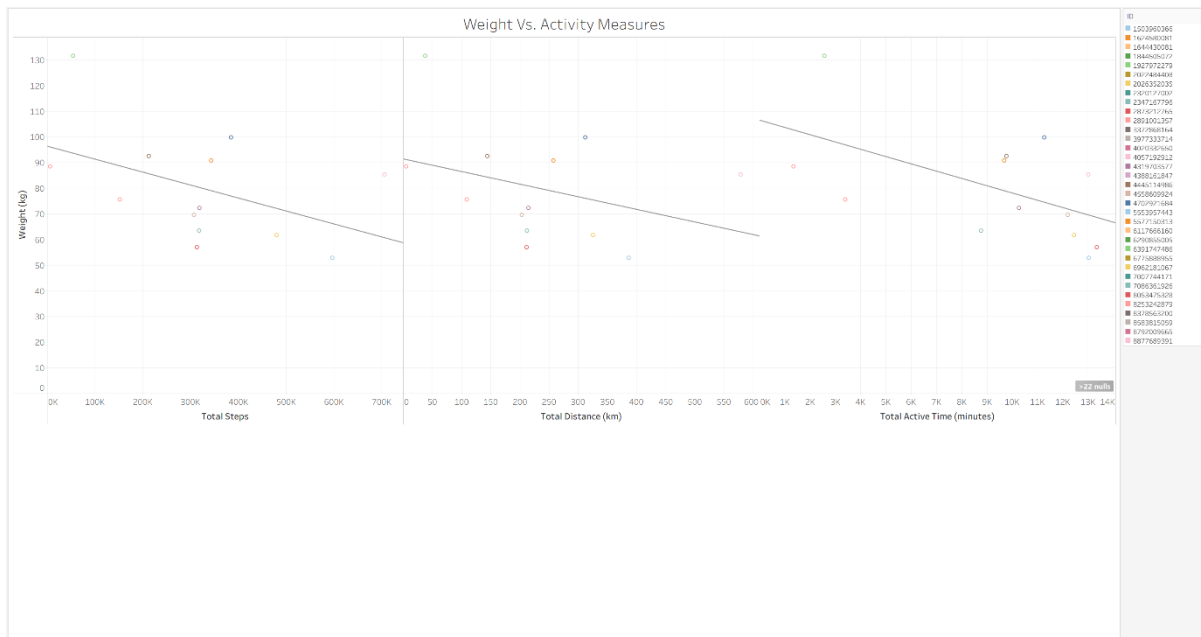


Figure 2: Weight vs. Activity Measures

“Weight vs. Activity Measures” illustrates the relationship between body weight and various activity metrics, including total steps, total distance, and total active minutes. Across all three visualisations, a negative association is observed between individual users’ body weight and activity levels, suggesting that higher activity levels are associated with lower body weight. However, the wide dispersion of data points indicates that this relationship is weak.

Additionally, the visualisations suggest potential differences in product usage patterns between higher-weight and lower-weight users.

Implications

These findings support the segmentation of users to inform tailored marketing and product strategies that address differing user needs and behaviours. Such segmentation has the potential to improve user satisfaction and support growth in the overall user base.

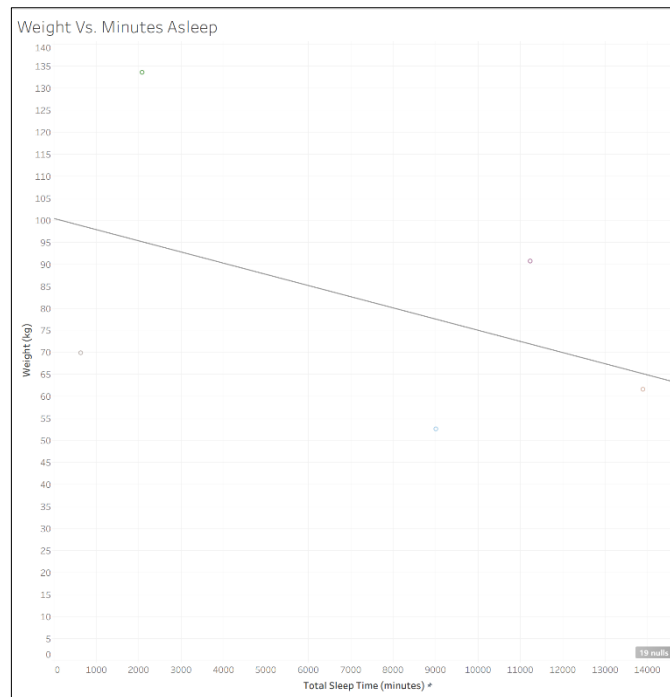


Figure 3: Weight vs. Time Asleep

“Weight vs. Time Asleep” illustrates the relationship between body weight and sleep duration. A negative association is observed between individual users’ body weight and sleep duration, suggesting that longer sleep duration is associated with a lower body weight. However, the wide dispersion of data points indicates that this relationship is weak.

Implications

These findings further support the segmentation of users to inform tailored marketing and product strategies that address differing user needs and behaviours.

6.0 Recommendations

Based on the analysis of user behaviour and engagement data, the following recommendations are proposed for the Bellabeat marketing team:

1.

Segment Users Based on Activity and Health Engagement

Insight:

Analysis indicates significant variation in activity levels and health outcomes across the user base. While some users demonstrate consistent engagement and higher activity levels, others show minimal changes in weight and health metrics over the 90-day period.

Recommendation:

Bellabeat should categorise users into distinct segments based on activity intensity and engagement patterns, such as:

- Highly active users
- Moderately active users
- Low-activity users

This segmentation would enable more targeted product experiences and communications, improving relevance and engagement for each group.

2.

Develop Tailored Features to Encourage Health Outcomes

Insight:

Despite using a fitness product, many users showed little to no change in key health indicators such as weight, BMI, or resting heart rate, suggesting that current features may not sufficiently motivate behavioural change for all users.

Recommendation:

Bellabeat should introduce segment-specific features, including:

- Motivational prompts and achievable goal-setting for low-activity users
- Progressive challenges and milestone rewards for moderately active users
- Advanced performance insights and achievement tracking for highly active users

These tailored features may help drive sustained engagement and encourage measurable health improvements across different user segments.

3.

Align Marketing Messaging with Diverse User Motivations

Insight:

The lack of measurable health changes among some users suggests that not all consumers engage with Bellabeat products primarily for health improvement. Alternative motivations, such as aesthetic goals or identity reinforcement, may influence usage.

Recommendation:

Marketing strategies should reflect varied user motivations by:

- Promoting lifestyle, wellness identity, and aesthetic benefits alongside health outcomes
- Highlighting personalisation and habit-building rather than weight loss alone
- Positioning Bellabeat as a holistic wellness companion rather than solely a fitness tracker

This broader positioning may resonate with a wider audience and support growth in user adoption.

7.0 Conclusion

This analysis examined user activity and health data over a 90-day period to identify patterns in engagement, health outcomes, and product usage among Bellabeat users. Overall, the findings indicated that while users demonstrate varying levels of activity, these behaviours do not consistently translate into measurable improvements in key health metrics such as weight, BMI, or resting heart rate.

Furthermore, observed differences in engagement across users suggest that Bellabeat's user base is not homogeneous in either motivation or usage patterns. Some users may engage with the product for reasons beyond measurable health improvement, including lifestyle, aesthetic, or identity-related factors.

Collectively, these insights highlight the opportunity for Bellabeat to better segment its users and deliver more personalised product features and marketing strategies. By aligning functionality and messaging with diverse user needs and motivations, Bellabeat can improve user engagement, enhance satisfaction, and support sustained growth in its customer base.

8.0 Appendix

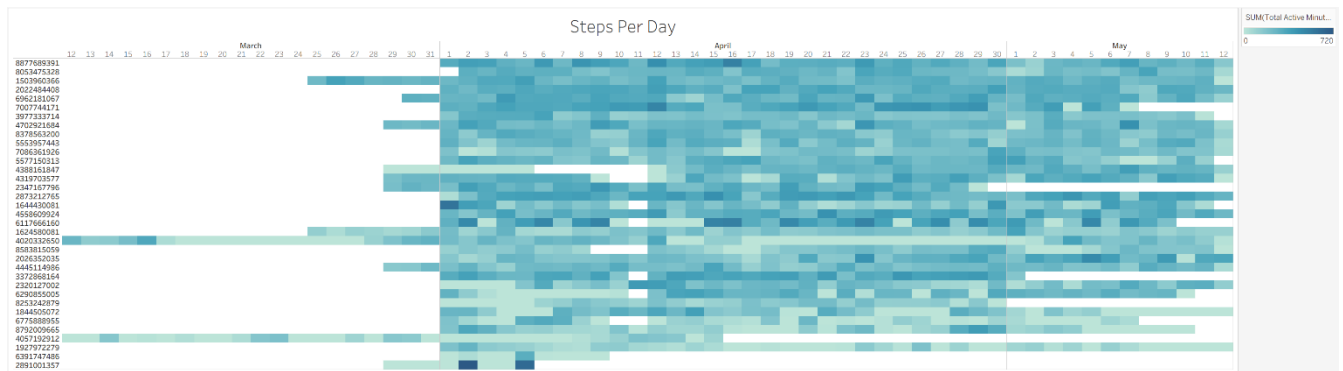


Figure 6: Steps Per Day

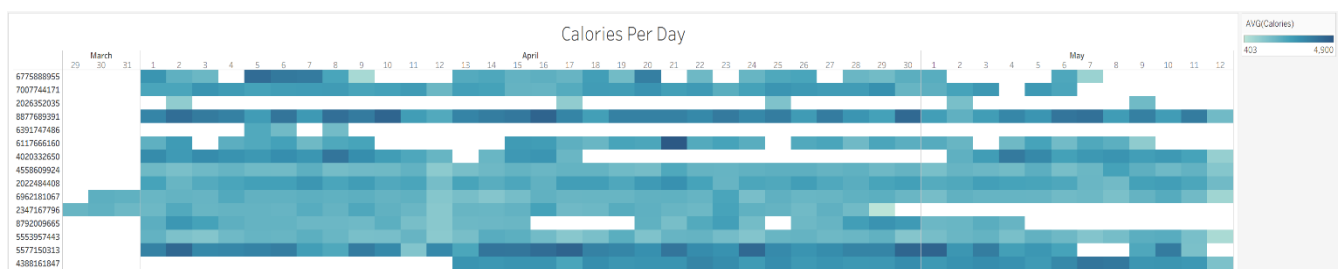


Figure 4: Calories Per Day

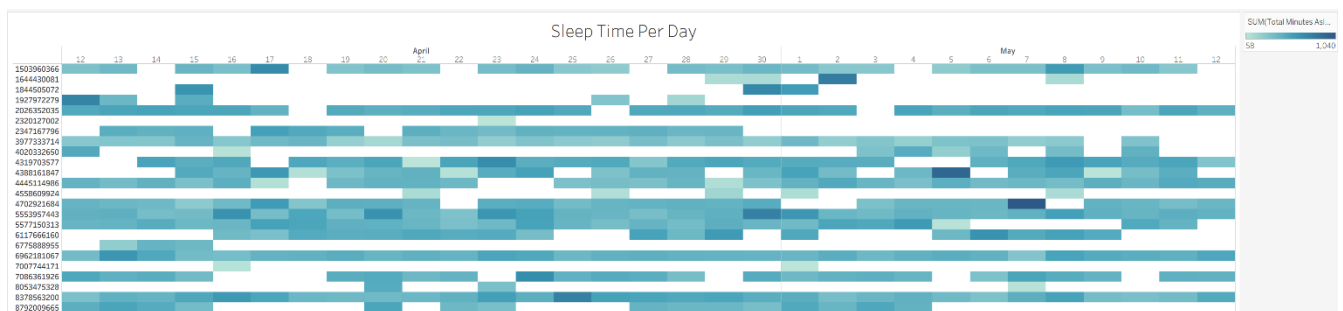


Figure 5: Sleep Per Day

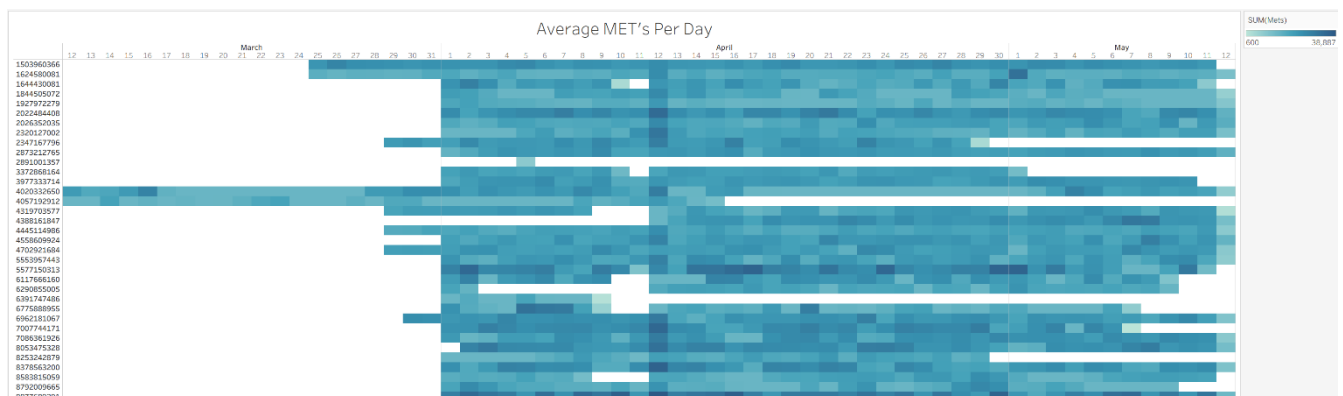


Figure 7: Average MET's Per Day

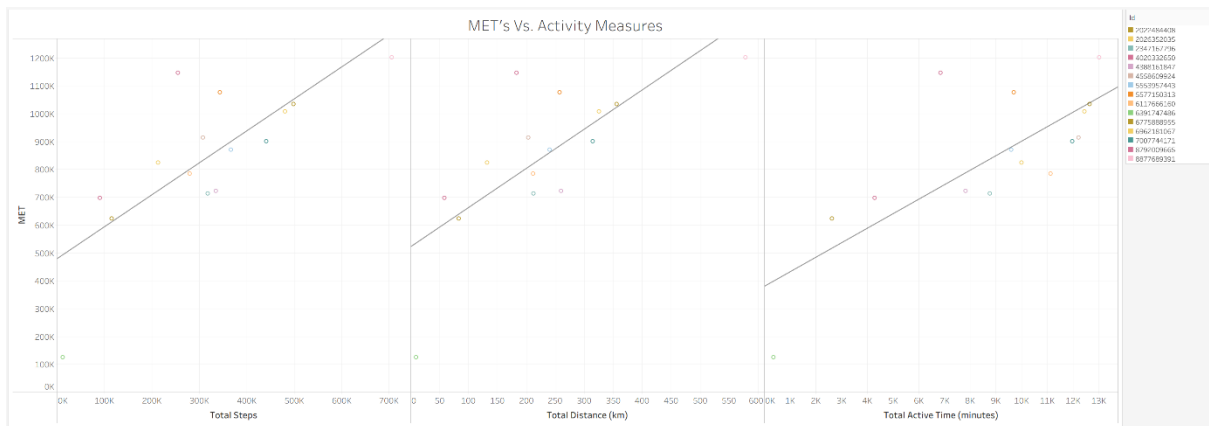


Figure 9: MET's Vs. Activity Measures

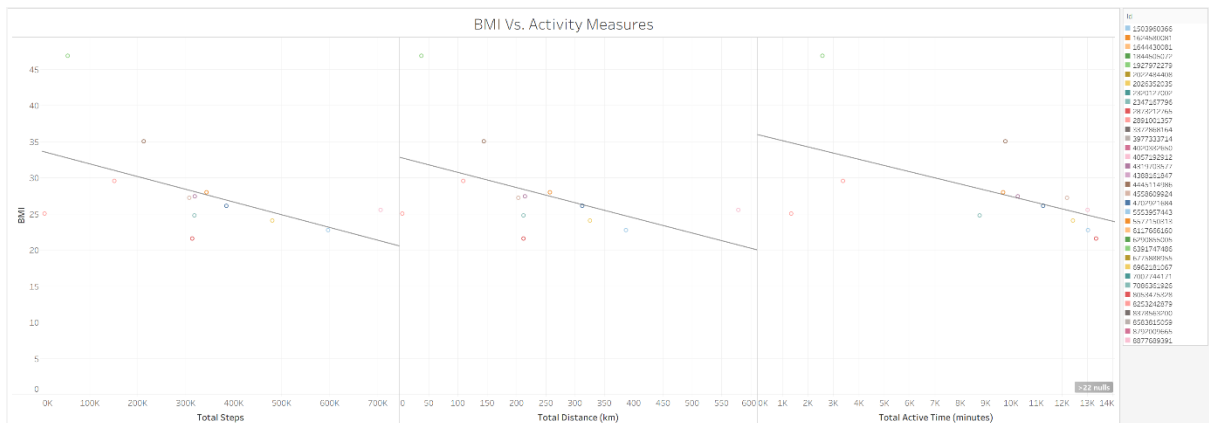


Figure 8: BMI Vs. Activity Measures



Figure 10: Sleep Vs. Activity Measures

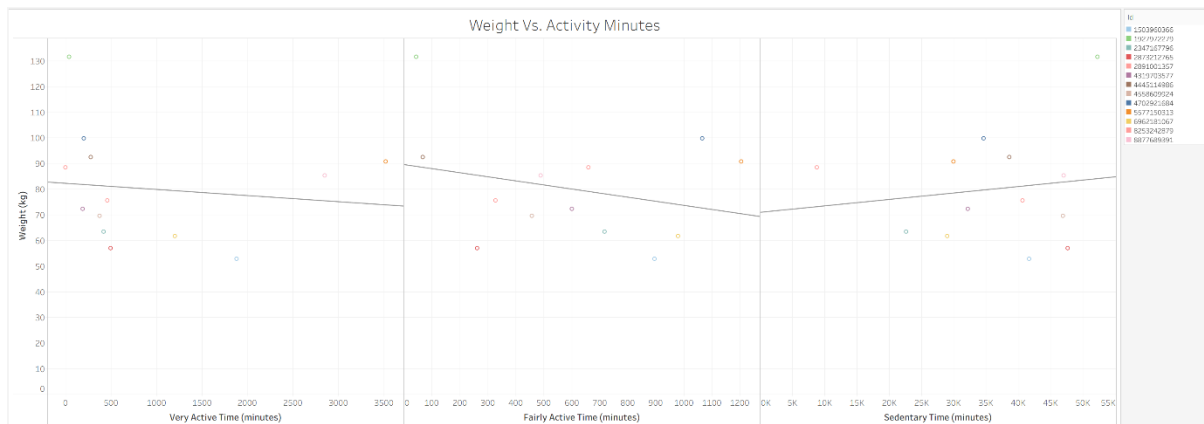


Figure 12: Weight Vs. Activity Minutes

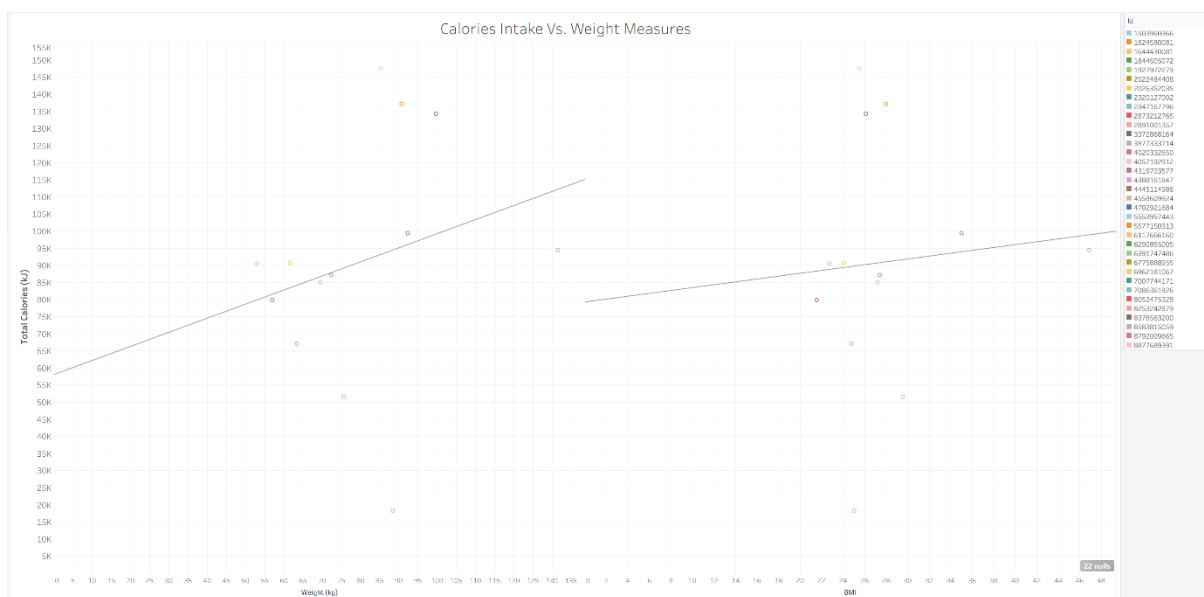


Figure 11: Calories Vs. Weight Measures

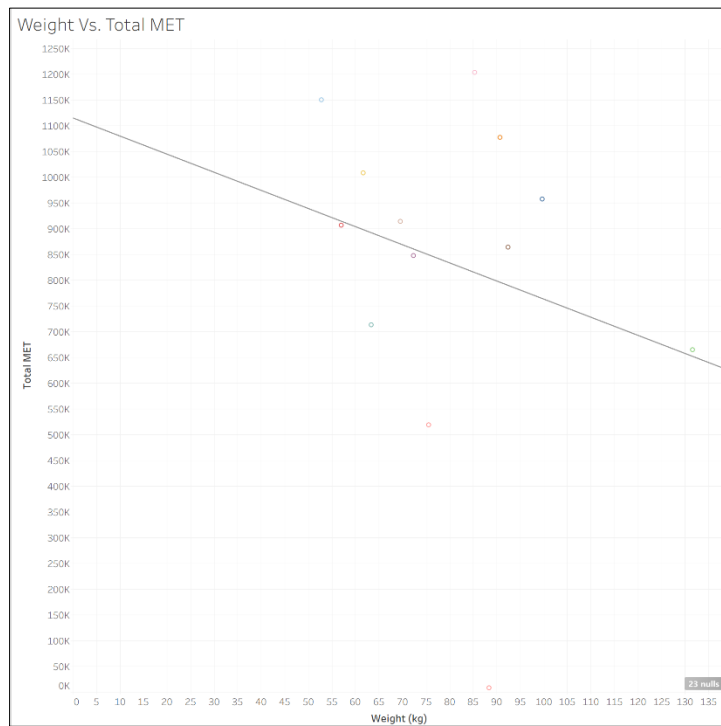


Figure 13: Weight Vs. Total MET

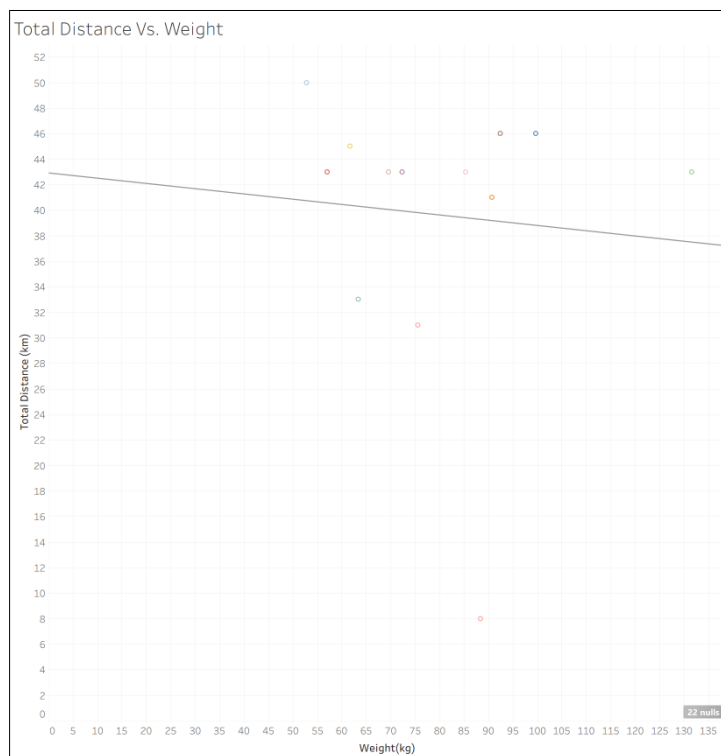


Figure 14: Weight Vs. Total Distance